

Syllabus

Spring semester 2026

Course Title	General Earth Science Lab		Course Code	PE3204	Credit	1	Classroom	TAMGU 2405
			Type	Experiments	Hours	2		
Teacher	Name	Class	Office	Office hours	Contact		Classroom code	
	Seo Mira	1	TAMGU 2401	Mon. 16:30-18:30	mrseo165@gmail.com (email only)		uvz5zzwg	
		2		Tue. 15:30-17:00			i75mmhvv	
		3					k7lbkuro	

1. Course Description

This course focuses on the physical processes operating in the hydrosphere, geosphere, atmosphere and biosphere of the Earth. This course covers the Earth's internal processes with topics including plate tectonics, minerals, rocks and earthquakes. We explore the Earth's atmospheric processes and meteorological phenomena and the dynamics of the oceanic and biological systems. We describe the scientific method and apply it to experiments presented in the lab to explore the dynamics of the Earth's subsystems. Also we work cooperatively in a group to successfully complete lab experiments, an results, discuss questions, and state conclusions.

2. Core Target Competencies

Subject Core Competencies						
. Analytical and Creative Thinking	Adaptability/Progression/ Spirit of Adventure	Integrity and Ethical Decision Making	. Leadership and Teamwork	. Complex Communication	Digital/ Quantitative Literacy	Habits/ Mindset

3. Textbooks and References

A. Textbook: Developed teaching materials

B. References

- Geology from Experience: Hands-on Labs and Problems in Physical Geology, E. Kirsten Peters and Larry E. Davis, W.H. Freeman and Company.

4. Class Method and Evaluation

A. Evaluation Criteria

Report	Presentation	Attitude	Attendance	Total(%)
80	5	5	10	100

B. Evaluation Method

(1) **Lab report 80%** : pre- and result reports (Individual evaluation):

- The report file format should be PDF or jpg, as possible.
 - In time you will get the score range of A to D, and it will normally be 1-2 days, or the exact deadline will be announced in the classroom.
 - The later your report submission, the more points you will lose. If you submit your report after 1-2 days and within 1 weeks, you will receive a score ranging from B to F. Nevertheless, submitting the report, even if late, will help your score more than not submitting it at all.
 - If you have any problems, please consult with the teacher in advance.
- * **Plagiarism**; A meeting of the faculty committee will be open for the FAIL of the course.

- Is there an accurate understanding of the purpose of the experiment?
- Is creative answer in the questions?
- Is there any further investigation?
- Make good use of figures and tables?

(2) **Presentation 5%**: (Group/Individual evaluation)

- Pre-report and results for full reports..
- Randomly 2-3 people at each experiment

(3) **Attitude 5%**: (Individual evaluation)

- Organization; Quality of information; Interest level and Neatness.
- Experiment progresses; Laboratory safety instructions.
- Cooperation; in team activity, sincerity, honesty, fairness, and respect.

(4) **Attendance 10%**: (Individual evaluation)

- Unexcused absence makes the score degraded by 20% each time.
- If 4 times of unexcused absence, a meeting of the faculty committee will be open for FAIL of the course.

5. Criteria for Awarding Credits

☞ Pass: In the case of receiving a **score of 80 or more** out of 100 according to the criteria of the

evaluation method.

- ☞ **Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.**

- Attendance for at least three-quarters of the total class hours is required to earn credit.
- For Pass/Fail courses, if the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.
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- Calculation of the basis score for awarding credit to absentees :
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- If a basis score cannot be calculated, the grading method :
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- ※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

7. Weekly Plans

Week (period)	Topic and Contents	Concepts for pre-lab	(Notes)
Week 1 2.23.~3.1.	Introduction to the syllabus		(2.23 Mon. Entrance ceremony)
Core Competencies	-Complex Communication		
Week 2 3.2.~3.8.	Understanding the Earth System (1)	-Earth system - Earth's subsystems	GA Presentation Full Report (3.2. Mon. Substitute Holiday)
Core Competencies	-Analytical and Creative Thinking -Complex Communication -Global Perspective -Integrity and Ethical Decision Making	and their interactions	
Week 3 3.9.~3.15.	Presentation of the Earth System (2)		
Core Competencies	-Analytical and Creative Thinking -Complex Communication -Global Perspective -Integrity and Ethical Decision Making		
Week 4 3.16.~3.22	Driving force that moves the plate	-Plate tectonics -Three boundaries of planets	Worksheet
Core Competencies	-Analytical and Creative Thinking -Digital/ Quantitative Literacy -Adaptability/Progression/ Spirit of Adventure		
Week 5 3.23.~3.29.	The movement of the plate using a GNSS Time Series	-Pangaea -Tectonic cycle	Worksheet
Core Competencies	-Analytical and Creative Thinking -Digital/ Quantitative Literacy -Adaptability/Progression/ Spirit of Adventure		
Week 6 3.30.~4.5.	Climate action net zero	-global warming -carbon neutrality	GA Worksheet
Core Competencies	-Global Perspective -Complex Communication		
Week 7 4.6.~4.12	Determine Location of an Earthquake	- Seismic Waves : P and S wave	Worksheet
Core Competencies	- Analytical and Creative Thinking - Global Perspective - Adaptability/Progression/ Spirit of Adventure	- Locating an earthquake	
Week 8 4.13.~4.19.	Midterm Exams (4.14.~4.17.)		4.13. Regular Classes
Week 9 4.20.~4.26.	IRIS Earthquake Browser	-Magnitude scales of Earthquakes	Worksheet
Core Competencies	- Analytical and Creative Thinking -Global Perspective - Adaptability/Progression/ Spirit of Adventure		

Week (period)	Topic and Contents	Concepts for pre-lab	(Notes)
Week 10 4.27.~5.3.	Identifying sedimentary rocks, igneous rocks and metamorphic rocks (1)	-Three type of rocks -Properties of rocks	GA Worksheet 5.5. Children's Day
Core Competencies	Analytical and Creative Thinking Complex Communication		
Week 11 5.4.~5.10.	Identifying sedimentary rocks, igneous rocks and metamorphic rocks (2)		
Core Competencies	Analytical and Creative Thinking Complex Communication		
Week 12 5.11.~5.17	Water temperature variation with depth	- Vertical Stratification of the Ocean	GA, Worksheet
Core Competencies	Analytical and Creative Thinking Leadership and Teamwork		
Week 13 5.18.~5.24	Thermocline of real ocean	- Seas around the Korean Peninsula - Major Water Masses	Worksheet
Core Competencies	Analytical and Creative Thinking Global Perspective		
Week 14 5.25.~5.31.	Greenhouse effect	- structure of the atmosphere -Greenhouse effect	Worksheet 5.25. Substitute Holiday
Core Competencies	Analytical and Creative Thinking Digital/ Quantitative Literacy		
Week 15 6.1.~6.7.	Introduction of the HYSPLIT		GA Presentation Full Report
Core Competencies	Analytical and Creative Thinking Global Perspective Adaptability/Progression/ Spirit of Adventure Integrity and Ethical Decision Making Leadership and Teamwork		
Week 16 6.8.~6.14.	Presentation free topic of the HYSPLIT (Make-up class week)		
Core Competencies	Analytical and Creative Thinking Adaptability/Progression/ Spirit of Adventure Integrity and Ethical Decision Making Leadership and Teamwork		
Week 17	Final Exams (No LABS)		

※ The above plans may change according to the academic schedule.

How to write an effective report

Almost IMRAD format:

Title - The title should reflect the factual content of the paper. Here you will write your name and the date on which the experiment was performed or the date the lab report was written.

(Abstract) – **Summary of the basic content of the experiment.** It should state the purpose of the experiment, mention the techniques used, report results obtained, and give conclusions. The point of the abstract is to give a concise summary of the whole report.

Introduction or Purpose – **It should provide the reader with information necessary to understand the scientific basis of the experiment and the techniques used.** In most cases, you should include background information and explain the theory behind the techniques. The most common mistake is that students make references. When you refer to any resources from others, you should **note** the source of the information. You are not supposed to write an introduction over the amount of half of the report. Keep in mind to write concisely key information and key technique.

Methods and Material –It should be included all raw data, observations, calculations, and appropriate graphs.

Results and Analysis – A summary of the final data or output. A short paragraph should be sufficient, along with any relevant charts and graphs labeled well. Remember to title and provide legends for all graphs and tables. The graphs and tables should be comprehensible independently of their association with the text.

Discussion or Summary - Discuss the experiment and the results obtained. This does not mean you simply report the results again, but rather **describe the key findings from the results, and interpret and discuss their significance.** Discuss whether your interpretations support or refute your hypothesis. **Discuss the reasons for your findings,** especially the main findings. Compare and contrast your findings with other studies. Present the strengths and weaknesses of your study including limitations. Suggest further research opportunities. **Make a final summary and conclusion.** Clearly state your overall findings and their implications.

Reference Citations - As required in all scientific literature, statements of fact, not considered "common" knowledge, must be properly referenced.

- Note 1: This kind of IMRAD format is a guideline that helps you to write a report. You don't have to follow this scheme, but you can re-organize the report depending on the topic t.
- Note 2: This guideline is adapted from one in the Writing Center, the University of Wisconsin, and modified for this class suitably